Power BI

Data Analyst	Business Analyst.
• It analyze data & find insights.	• BI works on these insights → To form strategy.
• They generally walk with data + technical skills. (SQL, Stats., PowerBI, Python, Hypothesis testing)	• They have non-tech skills. - Problem solving - Critical thinkers. - Good in communication. - Presentation skills. - Basic Tech. Skills (Excel, SQL) - Business / Domain knowledge.
• Workflow :— → Gather Data → Clean → Analyse → Insights.	→ Understand the problem → data → insights → Decision.

Supply Chain & Management :—

Cadbury
↓
insights on prev. Data. → Data Analyst
↓
forecast.

A : 10k ± 2k

B : 20k ± 3k

C : 10k ± 1k

D : 50k. ± 4k

$\xrightarrow{\text{Data given to}}$ Company → BA
↓

Production → Supply [Channel] → Handling Range (eg. ± 2k)

FMCG (Fast Moving Consumer Goods) :—

Trend

South India → Early & Mid June / Oct.

North India → July & Aug.

Central India → Early July & Mid Sept.

Sales get down.

[Marketing]

↓

Top notch.

consultancy

↓

KPMG

EY

Accenture.

Monsoon → Product + Monsoon Prone

MonsoonProne → Silica

↓

(Product Cost ↑) → BA

- Which product should be provided with silica.
- When to use silica. etc.

Power BI :-

(Business Intelligence)

- Power BI → Business Intelligence Tool. [Microsoft]

  Business Intelligence → Intelligently manage your business through technology.

Power BI	Tableau.
• Microsoft	• Salesforce
• Data Visualization. → 47 default * Constantly ↑	• 24 default. * No update.
• Advance Viz. → Third Party. → 528 viz. ≅ 580 – 590 charts. ↑ Charts → ↑ Inference	• Adv. viz. → 50 ≅ 60 → 100 ≅ 120 Viz.
• Data Sources ↳ 185 data sources (.csv, .xml, excel, api etc.) [Few Clicks]	• Data Sources ↳ 80 ≅ 90 [Additional tools] [Complex]
• Microsoft Compatibility	• Compatibility → x.
• Power Query → Data Cleaning	• Tableau Prep → Data Cleaning. [3-4y old] → Before External tools.
• AI → (Decomposition Tree, Analyze etc.) (Smooth)	• Smooth → x
• Windows.	• Mac + Windows
• User Interface → x	• User interface → ✓
• Updates → Regular	• Update → x.

Power BI	Tableau.
• 100TB {Storage}	• Unlimited.
• Processing → 15–20% ↓	• Processing → 20–30% ↑

Industry :–

Research [DA+BA]

1. Problem Identification. → Opportunities | Problems.

BA

2. Initial Consultation → Problem Content.
3. Problem Deeply Drive → Deep Study → Factors → internal [companies] / → external. [Society, govt.] } what why how etc.
   → In details.
4. Data Strategy → Data → ??
   (where to bring data from)

DA

5. Data Modelling → • Data Collection
   • Data Preprocessing.
   • Data Cleaning.
   • Data Transformation
   • Data Visualization

BA

6. Insights → findings
7. Decisions.

Power BI.

Power BI Components :-

Power BI Desktop (50%)	Power BI Services (50%)	Power BI Mobile.
• Windows Software ↳ - Dashboards - Load data - Data clean - Graph - Viz. - Report generations. • Power Query.	• Online Cloud ↳ - Online Services. • Diff. users → consumer. • Report Sharing. • Self-Service BI [non-tech] • Data Refresh • Security Measures. • AI features.	• Viewing & Interactions. • Notifications.

Excel $\xrightarrow{\text{to}}$ Power BI

- Data was recorded manually.
- Excel was able to handle it.

- Now every business is data centric.
- When we load that amount of data in excel, it crashed.
- People moved to data lakes, data warehouses etc.
- People moved to PowerBI. → BI Tools.

Power BI Dataflow :-

Data Collection → Power BI Desktop. → Power BI Services. → Client.

- Create Visuals.
- Data formatting.
- Data Cleaning.
- DAX, M
- Power Query.

⟹

Cloud Enabling :-

- Refresh
- Row level Security.
- Dashboard
- Reports.
- Self Service BI.

Data Collection. :-

3 forms →
- Structured Data → Rows & Cols. [Tabular].
- Unstructured Data. → Text, Video, Images, ...
- Semi-structured Data. → Mix of Both, JSON, XML.

eg.) Semi-structured Data.

4" 5" 6"

Cust. A → 4" pipes. ✓
Cust. B → 4" + Brand name ✗

} thus. → Semistructured

Data → from where it is coming & how.??

Data → Databases, Datawarehouses, Datamarts, APIs

Store to

↓

Processes : ETL, Webscrapping

} Data Processing & Management

we do it with

↓

OLAP
OLTP

Databases :-

- Structured collection of data generally in tabular form.
  ↓
  [Rows & Columns]

- Transactions [day to day] frequent ; basic query.
  ↳ Current Data
  ↓
  - C → Create
  - R → Read
  - U → Update
  - D → Delete.

- Large no. of frequent & basic queries → process.

Data Warehouse :-

- It is a centralised repository for storing data from multiple sources.
- Current + Historical Data stored.
- Used for analytical, BI, strategy purposes.
- It is designed to handle complex queries.
  ↓
  - Aggregations
  - Joins
  - Mergers.

Data Mart :-

- A subset of data warehouse, that stores information about a particular subject.

  eg.) Company → Database → Finance, HR, Marketing.

Data Lakes :-

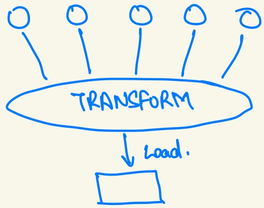
- Structured, Unstructured & Semi-structured data resides in it.

API's → link to fetch data from server.

How data is collected?

used in industries.

1. ETL ⟶ E → Extract [csv, API, unstructured]
   T → Transform [convert ⟶ needed]
   L → Load. [Datawarehouses, Databases]



2. Web Scrapping → Website, data Scrap
   ↓ convert to
   Structured form.

3. Manual Entries.

4. IOT Devices (Internet of Things) Devices.

OLAP & OLTP :-

OLAP, OLTP ↓ **Conceptual framework** [Data processing & management]

OLTP: Online Transactional Processing.

- Handles routines, day to day ops.
- Transactional Tasks.
- Simple, straightforward queries [CRUD]
- It stores current data, Operational data.
  ↓ designed
  Large vol. of small trans. handle.
- **It prioritizes low-latency response.**

eg. ATM Concept.

OLAP: Online Analytical Processing.

- Complex queries & multidimensional analysis.
  ↓
  - BI, data analytics, etc.
- Complex queries ⟶ handles easily.
  [aggregations, calculations, logics, merge, joins etc.]
- Historical data stores + Current data feed.
  Data warehousing.
- **Response time is slow.**
  ↳ * Data → Correct
  * As per requirement.

eg. PowerBI concept.

Power BI Desktop

- ↳ Data Cleaning.
- ↳ Data transformation
  [Grouping, joining, granularity
  ↓
  minority
  eg.) hourly basis.
  daily basis ↲ convert
- ↳ Data Visualization
  [Graphs]
  ↳ logic, data
  ↓
  $\boxed{\text{DAX, M-Language}}$ $\xrightarrow[\text{for}]{\text{Also used}}$ Data transformation & cleaning.
- ↳ We interact with these visuals to draw inferences.

- ↳ Online [PowerBI service]
  - → Publish.
  - → Refresh.
  - → Security.
  - → Admin Rights. → Changes.
  - → Self service BI

Client & BA team. ←

Slicer :-

* It's a type of visual, its always in report page view.
* Basic — Select OR Deselect.
* Can connect/disconnect a visual

Filters :-

* Functionality, (x) working Area, Separate Pane.
* Adv., logical statement, filteration.
* Applied to all.

Performance & Applicability :

* Filter > Slicer

Data Modelling :-

It's a process of creating a conceptual representation of data, its structure, & relationship b/w different data (tables).

It's like if you want multiple tables to communicate with each other you create a relationship b/w them.
This is called data modelling.

Data Model :-

Data model is a set of collection of tables, its attributes, relationships b/w diff. tables, calculations, DAX, measures or any other thing that are used to create visualization & perform data analysis.

Connect [talk]

Sales	Customer
Cx id	Cx id
Sales	Cx Country
Order id	Gender

Data Modelling

Connection → Common feature.

Sales by Gender → Incomplete (x)
Scattered.

Cardinality :-

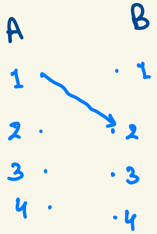
How 2 rows are connected b/w tables.

Types : One to One
One to Many
Many to One
Many to Many.

Sales	Customer
Cx id	Cx id
Sales	Cx Country
Order id	Gender

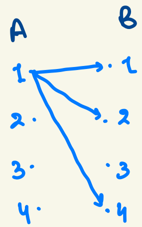
many. to 1

One to One

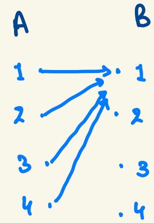


eg. Adhar & PAN card

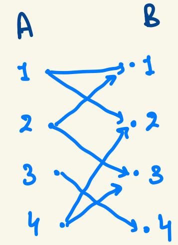
One to Many



Many to One



Many to Many



DAX (Data Analysis Expression) :

Adding Column.

→ Table → Table Tools → New Column

Col. Name — fx — table — table col.

Month = MONTH(Orders [order date])

Year = YEAR(Orders [order date])

Day = DAY (Orders [Order date])

Quarter = QUARTER(Orders [order date])

dayofweek = WEEKDAY (Orders [date])

By default PowerBI take

Sunday = 1
Mon = 2
⋮
Sat = 7

If we want Monday as 1
then
WEEKDAY (Orders [date], 2)

Format :-

formatting smthg. acc. to condition.

- Numeric
- Text
- Date

We can convert them to desired format.

DAX — table — col. name — keyword

Profit = FORMAT(Sheet1 [Profit], "Standard")

→ Google → DAX → learn.microsoft.com → $\boxed{\text{format}}$

DAX (Data Analysis Expression) :

FORMAT :-

Month Name = FORMAT(Orders[Order date], "mmm")

UPPER :-

Medium Caps = UPPER(Orders[Order Priority])
LOWER

Selected alphabets :-

Selc 1 = LEFT(Orders[order Priority], 3)
↳ gives 3 letter from left

RIGHT(Orders[order Priority], 3)

MID(Orders[Order Priority], 2, 4)
2 ↑ Start index
4 ↑ No of characters.

Q. The Client requires a visual rep. to display the status of order processing.
Orders will be classified as delayed if the shipping date exceeds the order date by more than 2 days.

Steps :-

- Diff.
- Diff > 2, delay, on-time.

Delay|On-Time = DATEDIFF(Orders[Order Date], Orders[Ship Date], DAY)

Delay|On-Time_2 = IF(Orders[Delay|On-Time] < 2, "On-Time", "Delay")

Orders[Delay|On-Time] < 2 ↓ Condition.
"On-Time" ↓ True
"Delay" ↓ False

DAX (Data Analysis Expression) :

Q.) Now client wants no. of orders delayed as well as were on-time.

- Pie - Chart.
- Legend → Delay / On-time_2
- Values → Order id

* Try to make less columns. Colums occupy memory. Thus, more expensive.

2. The client requires a detailed analysis to understand sales patterns based on the day of the week. They need a pie chart that visually compares the average sales volumes on weekends versus weekdays. This will help in identifying any significant differences in consumer behaviour and sales performance, enabling more informed strategic decisions for marketing and resource allocation.

Method 1 :-

```
Weekends/Weekdays = SWITCH(Orders[Day Name],
                        "Saturday", "Weekend",
                        "Sunday", "Weekend",
                        "Weekday")
```

Method 2 :- (Industry Approach)

```
Weekends/Weekdays = VAR days = FORMAT(Orders[Order Date], "dddd")
RETURN
IF(days IN {"Saturday", "Sunday"}, "Weekends", "Weekdays")
```

DAX (Data Analysis Expression) :

3. The company, CA, seeks a comprehensive analysis of sales data segmented by financial months and financial quarters. This breakdown will reveal sales performance trends and patterns within each fiscal period, facilitating effective strategic planning and accurate financial forecasting

Logic : months ⟶ Financial Months (Apr – Mar)

(month > 3) – 3

if (month < 3) + 9

4	1	Apr
5	2	May
6	3	June
7	4	July
8	5	Aug
9	6	Sept
10	7	Oct
11	8	Nov
12	9	Dec
1	10	Jan
2	11	Feb
3	12	March

DAX

Financial Month = VAR month = Month(Orders[Order date])

RETURN

"FM_" & IF(Month>3, month-3, month+9)

"FM_" ↓ Text   & ↓ Concat   IF(...) ↓ #number

O/P

FM-1

FM-2

Logic Quarters :

Apr, May, June – Q1

July, Aug, Sept – Q2

Oct, Nov, Dec – Q3

Jan, Feb, Mar – Q4

DAX (Data Analysis Expression):

4. The company, Finance Manager, requires an analysis to determine the average cost per unit, segmented by category and subcategory. This detailed breakdown should provide insights into cost distribution and highlight areas for potential cost optimization, aiding in more precise budgeting and pricing strategies.

   Cost Price = Sale Price – Profit – Shipping Cost

Cost/Unit = (Orders[Sales] - Orders[Profit] - Orders[Shipping Cost]) / Orders[Quantity]

- Make Matrix
  - Values → Orders[Cost/Unit]
    Average
  - Rows → Categories
    Subcategories

5. Management wants to cluster its customers in different age groups.

14-19	Teen
20-30	Young
30-40	Adult
40-50	Old Adult
>50	Old

After Clustering, companies want a visual that will show total numbers of consumers in each category and what is the contribution of each category to total consumer population.

```
Category = VAR Age = DATEDIFF(Customer[Date of Birth], DATE(2019, 12, 31), YEAR)
RETURN
SWITCH(TRUE(),
    Age <= 19, "Teen",
    Age >=19 && Age <=30, "Young",
    Age >=30 && Age <=40, "Adult",
    Age >=40 && Age <=50, "Old Adult",
    "Old")
```

logical and.

- Pie Chart.

DAX (Data Analysis Expression) :

6. Clients have given the requirement to categorise the Income into different groups.

0-30k	A
30K-60K	B
60K-100K	C
100K -150K	D
>150K	E

Now, he wants a visual that will find the numbers of orders placed as per age category and income category in different regions.

```
Income Category = VAR abc = Customer [Yearly Income]
RETURN
SWITCH (TRUE(),
        abc <= 30000 , "A" ,
        abc > 30000 && abc < 60000 , "B",
        abc > 60000 && abe <100000 , "C" ,
        abc > 10000 && abe < 150000 , "D",
        "E" )
```

- Matrix

  Rows → Customer [Age Category]
  Customer [Income Category]

  Columns → Order [Region]

  Values → Orders [OrderID]
  Count

DAX (Data Analysis Expression) :

7. The client wants a card that will show the sales from the date 15th April 2019 to 15th November 2019.

Condition → Date 15/4/19 to 15/11/19.

DAX : CALCULATE

Use a New Measure :↓

Desired Sales = CALCULATE (sum (Orders[sales]), FILTER (Orders, Orders [Orders Date] > DATE (2019, 4, 15)),
FILTER (Orders, Orders [Orders Date] > DATE (2019, 11, 15)))

DAX (Data Analysis Expression):

8. Global Super Store wants to value their top 10 customers for their loyalty towards the store.
   Loyalty will be calculated on the basis of CLV Score [CLV score is Customer Loyalty Value Score]

   CLV score is calculated: -
   CLV = (Average Order Value* Average Profit Margin)/ Purchase Frequency.

Avg. Order Value = total sales / total quantity.
total sales
total quantity.
Avg. profit margin = M.P – C.P
M.P = sales / quantity (1 – discount)
Purchase frequency.

```
Profit Margin = VAR mp = Orders[Sales] / Orders[Quantity] *(1 - Orders[Discount]))
RETURN
mp - Orders[Cost]
```

```
CLV = VAR cust = Orders[Customer ID]

VAR sale = CALCULATE(SUM(Order[sales]), FILTER(Orders, Orders[Customer ID] = cust))
VAR quant = CALCULATE(SUM(Order[Quantity]  ), FILTER(Orders, Orders[Customer ID] = cust))
VAR profit = CALCULATE(AVERAGE(Order[Profit Margins]), FILTER(Orders, Orders[Customer ID] = cust))
VAR purchase = CALCULATE(DISTINCTCOUNT(Order[Order ID] ), FILTER(Orders, Orders[Customer ID] = cus ))

VAR avg_order_value = sale / quant

RETURN
(avg_order_value * profit) / purchase
```

- Apply filtering → top N → 10
  on customer Name

DAX (Data Analysis Expression) :

9. The Global Super Store Marketing team is interested in understanding how individuals of a specific gender, earning either above or below the average income for their gender, prefer ordering different subcategories across various regions. Based on this information, the marketing team aims to develop a targeted strategy.

Customer $\xrightarrow{\text{New Col.}}$ Avg.

Average = VAR male = CALCULATE (AVERAGE (Customer [Yearly Income]), FILTER (Customer, Customer [Gender] = "M"))

VAR female = CALCULATE(AVERAGE (Customer [Yearly Income]), FILTER (Customer, Customer [Gender] = "F"))

RETURN

if (male > Customer [Yearly Income] && Customer [Gender] = "M", "Under Avg.",

if (female > Customer [Yearly Income] && Customer [Gender] = "F", "Under Avg",

"Over Avg.")

- Matrix

  Rows ⟶ Orders [Regions]
  Customer [Gender]
  Customer [Avg Income]

  Columns ⟶ Orders [Subcategory]

  Values ⟶ Order [Customer ID] ⇒ count.

Measures :—

- Calculation / Aggregation → - To build logic
  - Business Problem solve.

- DAX only.

(Most use case of DAX.)

** [Measures are aggregations].

Measures

Implicit Measure

- Aggregation that PowerBI does on its own.

Explicit Measures.

- That we as a user create from our end.

** • Measures aggregate dynamically based on the context of visual.

Column — DAX — Measure

Column
↓
Permanently resides
↓
Occupy space

use
Row by Row {Value}

Measure → Space Saving.
↳ Action → call
Dead.

use
Aggregation | total of a column.
eg. find avg
find sum etc.

eg:) Profit % → Subcategory.

$$\text{Profit \%} = \frac{\Sigma \text{Profit}}{\Sigma \text{Sales}} * 100$$

(See Measure slide)

Measures :—

1. The client requests an analysis of the count of high-value customers segmented by region. High-value orders are defined as those who have made a purchase exceeding $1,500 with a profit percentage of at least 25% for that specific order.Client also wants to know out of total orders of that region how much population is high value customers.

- Table → Regions
- New measures.
- High Value Orders = CALCULATE(COUNT(Orders [Orders ID] , FILTER (Orders, Order [Sales] >1500, FILTER(Orders , Orders [P%] >25))

High Value Orders % = ([High Value Orders] / COUNT (Orders [Order ID]))* 100

2. The client has requested an enhancement to the previously created sales visual (Desired Sales). They would like the visual to display the variance between the desired sales of 2019 and desired sales of 2018 or the same date values.

Desired Sales 2018 =

VAR twenty = CALCULATE (SUM (Orders[Sales]) , FILTER(Orders, Orders [Order Date]> DATE(2018, 4, 15), FILTER(Orders, Orders[Order Date] < DATE(2018, 11, 15)))

RETURN

(([Desired Sales] - twenty) / [Desired Sales])* 100

Measures :—

3. The client seeks an analysis of the average order value per customer. Furthermore, they request an identification of the top 15 customers based on the highest total order values.

Average Sales Value = VAR Purchase = CALCULATE (DISTINCT COUNT (Orders [Order ID))

RETURN

[Sales Measure] | Purchase.

- filter → Top N → 15

4. The client requests the creation of a visual representation of sales data with a specific focus on country rankings. This visual should highlight and display only the countries that fall between the 11th and 19th positions based on their sales figures. The goal is to provide a clear understanding of the sales performance of countries that are mid-ranked in this range.

__Logic__ : Rank → Sales → Show Rank 11 to 19 Country.

Step1. → DAX : RANKX , BLANK

Rank of Country = IF (RANKX (ALL (Orders [Country] , [Sales Measure]) > 10 &&
(RANKX (ALL (Orders [Country] , [Sales Measure]) < 20 ,
[Sales Measure] , BLANK())

Measures :—

DAX : SELECTEDVALUE

Selected value = SELECTEDVALUE (Orders [Order Date]. [Year])

5. The client desires a visual representation of the cumulative sales sum for each year. This visual should clearly illustrate the progressive accumulation of sales over time within each year.

Logic : 2016 → Sales 2016

2017 → Sales 2016 + 2017

2018 → Sales 2016 + 2017 + 2018

2019 → Sales 2016 + 2017 + 2018 + 2019.

Cumulative Sum = CALCULATE (SUM (orders[Sales]), Orders [Order date]. [Year] <= SELECTEDVALUE (Orders [Order Date]. [YEAR]))

6. The CEO has requested a report showing the average daily sales amount across all years (2016, 2017, 2018, and 2019).

Logic : total sales of year / total no. of days.

year → 1st date , last date of sale of that year.

Average Sales = VAR lowest = MIN (Orders [Order date])

VAR highest = MAX (Orders [Order date])

VAR numdays = DATEDIFF (lowest, highest, DAY)

RETURN

[Sales Measure] / numdays.

Measures :—

7. The client requests a summary of the total sales for each month in the year 2019, specifically focusing on the sales from the last two financial quarters. This detailed monthly and quarterly breakdown will aid in assessing the performance during different periods of the year.

Logic : Oct Nov Dec → Q3
Jan Feb Mar → Q4

Table ↓
- Month → Filtering → Basic → =
- Sum of Sales.

8. The client wishes to know the number of churned (→ leave) customers for the years 2016, 2017, and 2018. The CEO is now interested in knowing how far we need to go in customer retention services.

Logic : 2016 → Last purchased date · 2016
2017 → " " " 2017.
2018 → " " " 2018.
⇓
- Customer 1st purchased date & last purchased date.
⇓
- grouping

DAX: SUMMARIZE

Retain Table =
SUMMARIZE(Orders, Orders[Customer ID],
"First Purchase Date", YEAR(MIN(Orders[Order Date])),
"Last Purchase Date", YEAR(MAX(Orders[Order Date])))

```
CustomerCountByYear =
VAR year = SELECTEDVALUE(Orders[Order Date].[Year])

RETURN
IF( year = 2016,
    COUNTROWS(Filter('Retain Table','Retain Table'[Last Purchase Date] = year)),

    IF(year = 2017,
        COUNTROWS(Filter('Retain Table','Retain Table'[Last Purchase Date] = year)),

    COUNTROWS(Filter('Retain Table','Retain Table'[Last Purchase Date] = 2018))
    ))
```

Measures :—

9. The client requires a visual that displays the top-selling product in each country. This visual should highlight the most popular products in different regions and provide insights into regional preferences and market demand. It will be useful for targeted marketing and inventory management strategies.

Logic : for each region, product being sold, quantity.

— SUMMARIZE , Product & region.

```
Step1 = SUMMARIZE(Orders,
                  Orders[Region],
                  Orders[Product Name],
                  "Total", CALCULATE(SUM(Orders[Quantity]))
)
```

For every region — product — quantity (max) — Name.

DAX : TOPN → Returns top N rows of specified table.

MAXX → Largest numeric value / string for each row in a table.

```
Most Sold Products =

VAR TopProducts = TOPN(1,
                        SUMMARIZE(Orders,
                                  Orders[Region],
                                  Orders[Product Name],
                                  "Total", CALCULATE(SUM(Orders[Quantity]))
                                 ),[Total],DESC
)

RETURN

MAXX(TopProducts,Orders[Product Name])
```

Name. display

Filtered Context :-

data → displayed → content → filter.

Filtered Context → Clear Context | Fix Context | Relative Context.

Fixed Context :-

Parameter Value → fixed, static.

Q.) Sales of consumer for year 2018?

Sales → Segment (fix): Consumer; Year (fix): 2018

Consumer 2018 = CALCULATE(SUM(Orders[Sales]), FILTER(Orders, Orders[Segment] = "Consumer"), FILTER(Orders, Orders[Order Date].[Year] = 2018))

→ Make Slicer of Segment.

→ Click for Other segment → i.e. Corporate | Home Office.

O/P ⇒ Blank.

Relative Context :-

Comparing 1 value in relation to other.

Eg.) Student A — 80; Student B — 90

Q.) How much marks of Stu A deviates from stud. B.

Relative Content :

eg.) I) Increase in Sales % for each as compared to 2016 sales.

Logic : 1. Sales 2016
2. Compare each & every yr. sale with 2016.
3. Calc. % change.

(New Measure:)

Relative Content =

VAR Sales 2016 = CALCULATE ([Sales Measure], Orders [Order Date]. [Year] = 2016)

RETURN

IF (SELECTEDVALUE (Orders [Order Date]. [Year]) = 2016 , " Base Year", ([Sales Measure] - sales 2016) / Sales2016

→ Convert to %

→ Make Table → Year, Sum of Sales, Relative Content.

eg.) II) Year on Year growth.

Logic : 1. For each year, previous yr. sales &
current " ".

2. Calculate %.

(New Measure)

YOY growth = VAR prev_yrSale = CALCULATE ([Sales Measure], Orders [Order Date]. [Year] = SELECTEDVALUE (Orders [Order Date]. [Year] - 1)

RETURN

([Sales Measure] - prev_yrSale) / prev_yrSale

→ Convert to %

Clear Content :-

We just remove filter on any value.

Q.) Visual to show contribution of each month to total sales.

Logic : sales of that month / total sales.

DAX : ALL → It returns value by ignoring all the filters.

(New Measure)

Total sum = CALCULATE (SUM(Orders[Sales]), ALL(Orders))

(New Measure)

Months Contribution = [Sales Measure] / [total sum]

★ DAX : ALLSELECTED

→ It nullify filters of the visuals but interact with any filter that is outside the visual.

(New Measure)

Total Sum allselect = CALCULATE (SUM (Orders [Sales]), ALLSELETED (Orders))

(New Measure)

Total Sale Non interacting visuals = CALLULATE(SUM(Orders[Sales]), ALL(Orders))

Mail -7

1. Analyse the company's product portfolio to determine the top 10 products that contribute the most to overall profitability.

Logic : For each product → total profit made.
for each product divide the profit of particular product with total profit.

Product Contribution to Profit =
VAR tot = CALCULATE(SUM(Orders[Profit]), ALL(Orders))
RETURN
[Profit Measure]/tot

2. Determine the leading product category for each year by analysing annual sales data. This analysis will help identify trends and shifts in consumer preferences, enabling the company to make informed decisions on inventory management, marketing strategies, and product development for sustained growth.

Logic : for each year → Top category.

Step 1: 2016 → A Sales 100, B 200, C 300; 2017 → A, B, C (Sales Val.)

Step 2: Rank every category for every year.

Step 3: Every year → Rank-1 → print.

```
Category_Rank_Sales = IF(RANKX(ALL(Orders[Category]),[Sales
Measure])=1,[Sales Measure],BLANK())
```

3. For each year, identify the top 3 managers annually based on the total sales they generated. This analysis will provide insights into the most effective leaders in driving sales performance, allowing the company to recognize and reward top talent, and replicate successful strategies across different regions.

Matrix → Rows → People[Names]
Cols → Year
Val → Sales.

```
Rank Of People = IF(RANKX(ALL(People[Name]),[Sales Measure])<4,
RANKX(ALL(People[Name]),[Sales Measure]),BLANK())
```

4. Assess how much the sales value of the top-ranked country deviates from the sales values of other countries within the top 10. This analysis will highlight the performance gap between the leading country and its peers, offering insights into potential areas for growth and improvement in underperforming regions.

Logic : 1. find rank of all countries.
2. Sales value of top country (Rank -> 1)
3. Sales value extracted ⟶ compare rest 9 countries on it.

```
Sales difference from top =
VAR country_sales_rank = RANKX(ALL(Orders[Country]),[Sales Measure])
VAR top1sale = CALCULATE( MAXX(
                TOPN(1,ALL(Orders[Country]),[Sales Measure],DESC),
            [Sales Measure])
)
VAR currentcountrysale = [Sales Measure]
VAR salesdifference = currentcountrysale - top1sale
RETURN
  IF( country_sales_rank<11, salesdifference/top1sale, BLANK())
```

5. Evaluate the year-over-year growth in new customer acquisition and the percentage growth in the total customer base. This analysis will help the company understand the effectiveness of its customer acquisition strategies and track the expansion of its customer base over time, providing insights for future marketing and retention efforts.

Logic : Any particular yr. ⟶ Cx base = New Cx + Existing Cx − Churned Cx

	Churned	New	Cx base
2018 →	1	100	100 − 1 = 99
2019 →	5	120	(100 + 120) − 1 − 5
2020 →	9	190	(100+120+190) − 1 − 5 − 9

year → year ⟶ Cummulative Sum of new Cx
year → year ⟶ Cummulative Sum of churned Cx

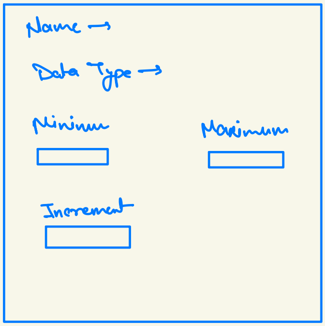
Cx Base

6. Client wants a visual where he will have authority to select the profit % and only the country that lies in that profit % range should be displayed.

Parametrized Charts :-

It's way of creating our own slicer.

Modeling → New Parameter → Numeric range



DAX: ISFILTERED → Returns True when the specified table or column is being filtered directly.

eg) Rank of Country = IF(ISFILTERED('Rank'[Rank]),   Condition

True

IF(RANKX(ALL(Orders[Country]), [Sales Measure]) < SELECTEDVALUE('Rank'[Rank]), [Sales Measure], BLANK()),

False

[Sales Measure])

Condition 2: If Client select 2 visual at same time.

DAX: HASONEFILTER → Returns True if filtered value on the column is one.

eg) Rank of Country 3 = IF(HASONEVALUE('Rank'[Rank]),

IF(ISFILTERED('Rank'[Rank]),   Condition

True

IF(RANKX(ALL(Orders[Country]), [Sales Measure]) < SELECTEDVALUE('Rank'[Rank]), [Sales Measure], BLANK()),

False

[Sales Measure]),

ERROR("Select only one value")),

[Sales Measure])

Parameterized Charts :-

Q. 6. Client wants a visual where he will have authority to select the profit % and only the country that lies in that profit % range should be displayed.

Profit % = ([Profit Measure] / [Sales Measure]) * 100

(Parameter)

Name → profit Slider

Min → -500

Max → 100

Increment → 1

Format → Slicer Setting → Style → Between.

```
Dynamic Profit =
VAR minimum = MIN('profit Slider'[profit Slider])
VAR maximum = MAX('profit Slider'[profit Slider])
RETURN
IF([Profit%]>=minimum && [Profit%]<= maximum, [Profit%],BLANK())
```

Power Query :-

→ Transformation can be done through power query
(like removing null value, duplicate value)

M- Language :-

→ Language generated / Code generated by the power query.
→ We should learn to read it.
→ View → Advance Editor.
→ We cannot use DAX in power query.
→ Only M Language is used.

Query Dependencies → Tells about the source of the data.
vs.
→ Data modelling = How 2 tables are related to each other.
↳ found in BI Desktop.

→ Table is called query in PowerBI.

Removing Null Value

Home → Remove Rows → Remove Blank Rows

Replace : (Right-Click) Replace Value.

null → Not Mentioned. : Convert Data type to text.
→ Replace to "Not Mentioned".

Add Column → Conditional Column : · We can only use if -else feature.
· Cannot use conditional operator (eg. OR, AND ..)

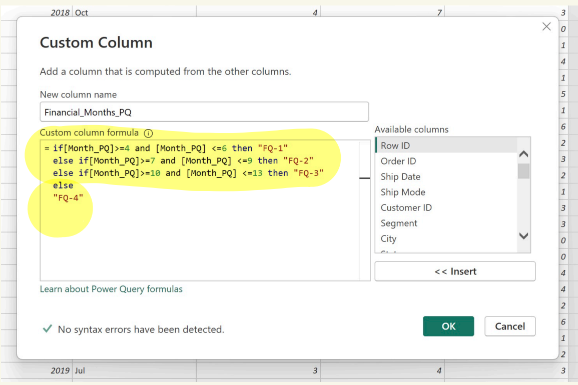
Home → Add Column → (Select Colum) → Standard → (Input no. to add).

Home → Add Column → Conditional Column.

New Column Name [ ]

	Col. Name	Operator	Value	Output
if	Day of Week	equals	7	Weekend
else if			1	Weekend
else	Weekday			

Custom Column : • Give us permission to create our own condition. Using M-Language.



Merging Columns.

(Select the Columns) → Transform → Merge Columns.

PowerBI asks for separator & Column Name

★ • We should try to save space with power query.
eg. Gender can be removed with Mr. & Mrs. infront of name.

Joins are called "Merge" in Power Query.

Home → Merge Query → (Select the tables)

→ (Select the join type)

A new row is formed, Click [▾], Select the column you want to use.

Replace "null" with "No"

Close & Apply.

Fuzzy matching → Used for matching similar type of data.

eg. M   Male

Similarity threshold

80

→ 80%

We can delete | unload the table.

(Right Click on table) → Uncheck [Enable Load]

* It will be shown in italics.

Append Queries → Joining 2 tables.

- Data types & Columns name should be same.

(Select the Sheet) → Home → Append Queries.

Sheet 1

Table to Append

Sheet 2.

↓

Ok.

Sheet 3 has an Additional Column, Append it to table.

→ It will append, the value for new columns → value remain null.

Mail - 8 :-

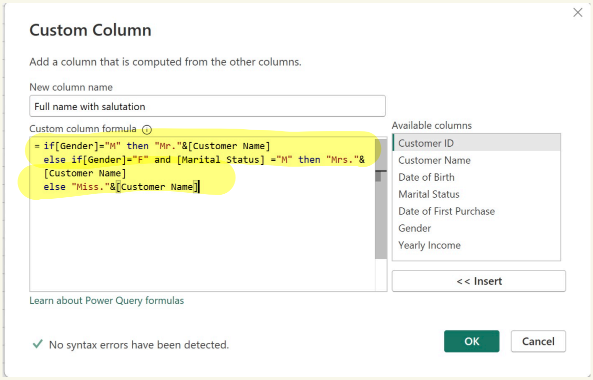
1. Find the total numbers of orders returned per market, and also analyse the returns for each market on the basis segment.

Table → • Markets → Filter → Returned
• Count of Returned • Yes

Bar Chart → X-Axis = Market
Y-Axis = Count of OrderID → Filter → Returned
Legend = Segment • Yes

2. Client wants to add the salutation before the names of Each customer, if the customer is male add Mr., if the customer is female and unmarried add Miss. Else add Mrs.

Transform Data → Customer Table.
Add Column → Custom Column



3. Show the number of companies, dealing in each category and subcategory, so that client can have an overview of the diversities of the company.

→ [Product Name] Column → Add Column → Extract → Text Before Deliminator → (Space) → OK ⇒ Company.

→ Matrix → Rows = Category
= Sub-Category
Values = Company ▸ Count

4. What is the percentage of quantity sold of routers, keyboards, headsets to total quantity sold in Accessories sub-category.

Logic : Rows → Keyboard, Router, Headset } words
Quantity sum.

(New Measure)

DAX : SEARCH → Returns the starting position of given text string.

```
Routers =
var  rout = CALCULATE(SUM(Orders[Quantity]),
                        FILTER(Orders,
                            (SEARCH("router", Orders[Product Name], 1 ,0)) >0))

var total = CALCULATE(SUM(Orders[Quantity]),FILTER(Orders,Orders[Sub-
Category]="Accessories"))

RETURN
(rout/total)*100
```

→ Make for keyboard & headset
→ Use Multi-Row Card

5. What is the percentage of quantity sold of leather Arm chair , rocking chair, adjustable chairs in Chairs sub-category.

6. What is the percentage of iPhone, Samsung, Motorola phones sold to total phones sold.

} Same Logic

Extra Question :-

Q.) Rank the region on the basis of the no. of sub-categories that are doing sales > 50000 and profit % > 10%.

And then rank the regions on the basis of the no. of sub-categories.

Matrix → • Regions.
• Sub-category
• profit %

(New Measure)

```
Qualified Categories = IF([Sales Measure]>50000 &&
[Profit%]>10,1,0)
```

```
Count of qualified sub-categories =
SUMX( SUMMARIZE(Orders,Orders[Region],Orders[Sub-Category],"Qualified" ,
[Qualified Categories]), [Qualified])
```

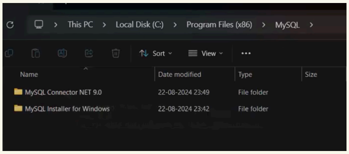
```
Region Rank = RANKX( ALL(Orders[Region]), [Count of qualified sub-
categories], ,DESC,Dense)
```

Keeps the ranks continuous.

Connecting SQL to Power BI.

Home → Get Data → $\boxed{\text{mysql}}$ → MySQL database → Connect.

* Download the connector → Net 9.0 on the popup link.
* Install it.

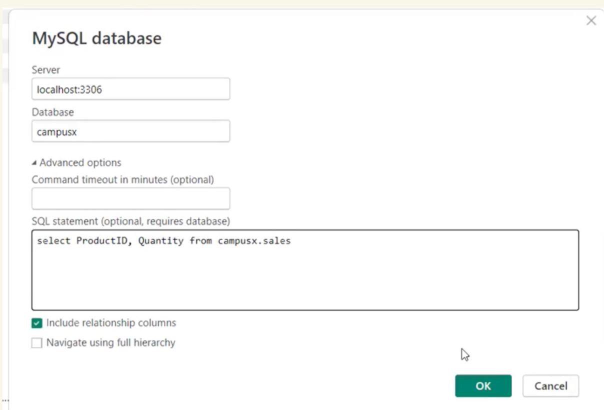


* Keep it in same folder.
* Restart the system.

Server → localhost / root.

Server → $\boxed{\text{localhost : 3306}}$
Database → $\boxed{\text{CampusX.}}$

* Note : We need the connector .



(for ORACLE)

* By default → Data is imported.

Connecting to MS SQL Server.

Home → SQL Server →

Server [ ]

Database [ ]

Data connectivity mode

- import → Download table to my local m/c from given server & DB. { used when dataset is big }
- Direct Query. → Work on the SQL server Not on local system. OR Live Data. { for small Dataset }

▾ Adv. options

Server → Left hand side → Right Click on Server
↓
Go to properties
↓
Copy Server name.

** imp

Query Folding :-

* Only applicable on DBs.
* It means we do the transformation in DB & then load it to BI desktop.

Home → Transform Data → (DB) → (Do the transformation at DB)

eg. we do like this (that we did in power Query) → Add Column → Custom Column → = [Sale Price] | [Quantity]

* Power BI send this automatically to server.

Go to step $\xrightarrow[\text{Click}]{\text{Right}}$ View Native Query.

**.
{ We have to keep all the steps together, that can be done in SQL }

{ Google → Query Folding → Blog by BIconnector }

{ MS SQL Server → Window → SQL Server Profiler → Connect }

{ Power query → Home → Refresh preview }

Python integration in PowerBI :-

{ PowerBI Desktop ⟶ Files ⟶ Option & Settings ⟶ Python Scripting. ⟶ (Set path.) }

{ cmd ⟶ pip install numpy
pip install pandas
pip install matplotlib
pip install seaborn }

Get Data ⟶ Python script ⟶ connect

Script

```
import numpy as np
import pandas as pd
data = pd.read_csv("Path")

df_manager = data[data['job_title'].str.contains("Manager", case=False)]
```

ok

⟶ Load.

Pivot & Unpivot :-

Pivot → Changing Rows to Columns.

{ (Select table) → Home → Pivot Column → $\boxed{\text{Value Column}}$ → OK }

Unpivot → Changing Cols. to Rows.

{(Select Cols.) → Home → Unpivot Columns}

Granularity :-

The depth at which we want to show the data.

e.g.) Orders table → Data is given in day wise

thus, granularity = Days.

if data is given in hours, then granularity = hours.

Which granularity we use is important.

e.g.) Client want data in monthly basis. Then,

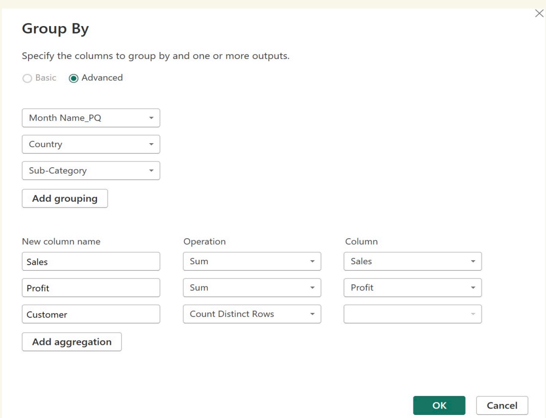
Orders table → Months, Country, Sub-category

Analysis → Sales, Profit, Count of Customer.

Soln.) { (Select table) → Home → Group By → $\boxed{\text{Basic → For 1 Column; Adv. → More than 1 Col.}}$ → OK }

(Select table): Transform Data

Here we group on basis of Month, Country, Subcategory.



Power BI Services :-

* Cloud based platform. | Central Repository.

Uses :-

- Report Publishing.
- Security setup.
- Manage Access.
- Reports.
- Share reports.
- [Apps:]* new.

* We need Organizational ID

{ app.powerbi.com } → BI Cloud Service.

1. Workspace → Create new workspace

{ PowerBI Desktop → Home → Publish }

Manage Access

↳ Workspace → Manage Access → Add Email.
(Select yours)

- Admin
- Member
- Contributor
- Viewer

PowerBI Roles
(Google it)

AD Group (Active Directory)

Group name

▾ Viewer.

{ Company Do it
Not on us. }

Power BI Services :—

Row Level Security (RLS) :— We control the row level data, not the visual.

*** imp : It applies to only viewers.

50% → is done in BI Desktop

50% → is done in BI Services.

BI Desktop → Modelling → Manage Roles → New (Name it) → (Select table) (Filter Data + New)

{ Make Rules }

Save

BI Service → Workspace (Select) ... (More options) → Type Semantic model → Add emails.

* Hidden Sheets are not shown on BI Service.

Enabling Map Visuals in BI Services.

{ BI Desktop → File → Options & Settings → Preview features → ☑ Shape Map Visual.

{ BI Services → Settings → Admin Portal → Integration Services (Enable All)

Power BI Services :—

(It gets enable)

Uses AI to make DAX.

BI Desktop → Quick Measures. → Suggestion With Co-piolet

↓ follow this step.

File → Options & Settings → Preview features → ☑ Q&A for live connected Analysis Service

☑ Quick Measure Suggestions.

Options & Settings ↓

Data Load → Q&A ☑ Turn on Q&A to ask natural language questions about your data.

BI Services → Settings → Admin Portal → $\boxed{\text{Quick Suggestion}}$ → (Enable All)

Sharing Report to End-User.

{BI Service → File → Embed report → website or portal.}

Another way ⤵
{ BI Service → Create App $\xrightarrow[\text{Next}]{\text{Add Content}}$ Publish app }

{ BI Service → Share }

↳ End user need to login.

Sharing Report to Client (as Viewer)

w/o logging in

→ Condition : There should not be Row level security (RLS)

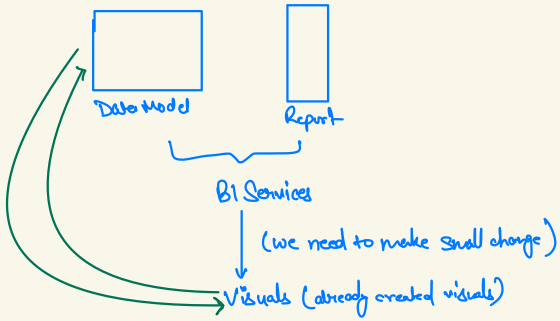
{ BI Service → File → Embed Report → Publish to Web }

in case of Error
{ Settings → Admin Portal → Export & Data Sharing (Enable All) }

Power BI Services :-

Self- Service BI :-

Allow client to make visuals on my data models.



Self Service BI

{ workspace (Choose yours) → New → Report → Pick a publish semantic model → Edit }

{* We get only already made table from that we can create visual. }

Power BI Services :—

Dashboard

* Dashboard is different from report.

{BI Service → New → Dashboard → (Name it) → Create}

Dashboard is a hyperlink to different-different reports.

{ Click on report → pin visual → Existing Dashboard → (Select Dashboard) → Pin}

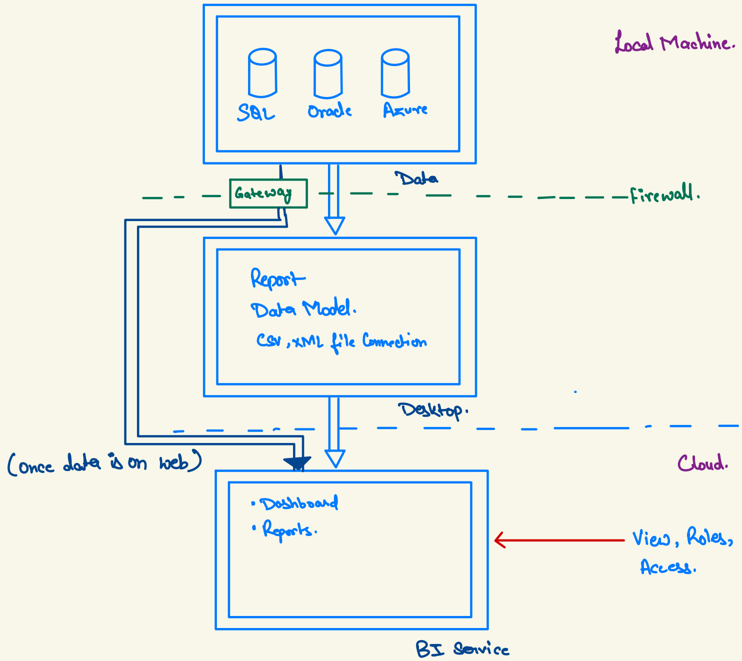
* We generally make dashboard & keep the most important information on dashboard.

Dashboard	Report.
• Non- interactive • Redirect to report. • Cannot create chart or visual. • Can pin only graph & visuals; but not slicers.	• interactive • Create Charts & models on already created data. • We can have visuals, graphs & slicers.

Power BI Architecture :-

Present at 2 places :

- BI Desktop
- BI Services.



Gateway

Standard Mode
(On premises gateway)
- Used in Organization ↴
  multiple source of data
  " users " "
- High Security.

Personal Mode.
(On-premises personal)
- On self system / One-system.
- Single source of data

Step 1

{ BI Services → Downloads → Data Gateways → Download (on-premise) (for single user) }

Step 2

{ → Sign-in (with MS fabric) → BI Services → Name ⋯ Semantic → Settings

→ Apply }

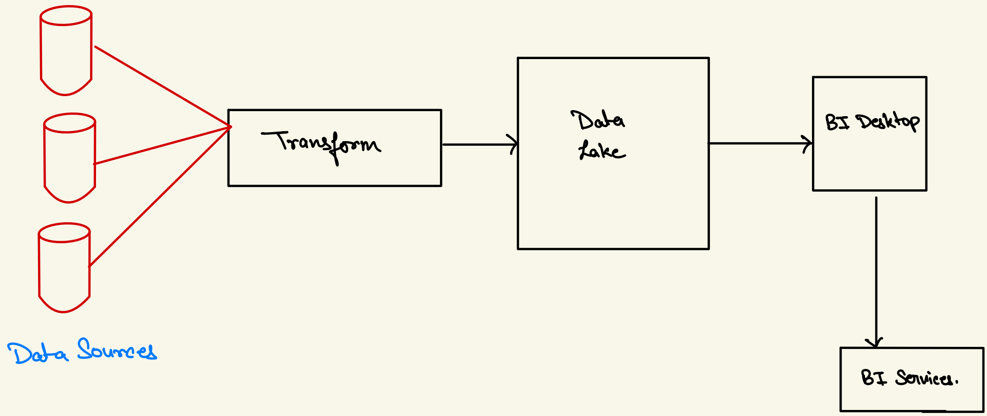
Connecting Gateway to Data Sources.↴

{ Edit Credentials → Privacy → none
→ SQL (if) → Window w/o impression → Privacy → none.
↓
Sign in. }

Refreshing data.↴

{ BI Services → Name ⋯ Semantic → Settings → Refresh → (Time Zone)
→ ON → (Select frequency & time) → (Refresh failure notification) (e-mail) → Apply.

Generally in industry



Power BI Pricing

vedantdobhal@BIservice19.onmicrosoft.com

1. Free → Validity [60 days] → Basic → 1GB.
   - → Reports
   - → Share.
   - → 8 times in 24hrs.

Developers:

2. Pro → 100GB (data model)
   - Data flow {many options}
   - 8 times refresh. in 24hrs.
   - → $10 user | month.

3. Premium

3(i) Premium Per User (PPU)
- 48x refresh | 24hrs
- Advance options
  - → Automation
  - → MS options.
- $20 user | month.

3(ii) Premium Per Capacity (PPC)
- Large Organizations.

10000 developers
- → 1000 developers : PPU → 10,000
- → 200 admins : PPU → 2000
- → 8800 viewers : Pro → 88000

Total cost ⇒ ≙ 1 lakh.

Just for viewers.
Unnecessary costing.

∴ Company goes for PPC
$5K | month.
- → Add many no. of viewers
- → 400GB Data.
- → 48 times.

Power Perform :-

Power Automate :-

Case 1 : BA Team $\xrightarrow{\text{Anytime}}$ Client
meeting.

- Should be ready with latest business data.
- Create a button for BA team.

BA click on button → gets updated .xls mail

{ Table → Build → Power Automate → (Sign in) → (follow Steps) }

{ (Click on visual) → Power Automate → ... → Edit → New → Instant Cloud flow → New Step →

→ (Search Select) → Select → Power BI Data (From) → (Enter key) (Enter Value) → New Step → (Send mail) —

→ (Send an email) → (To:) → Save → Save & Apply }

(Subj:)

(Body: Dynamic Content
↓
choose table)

{ Condition → Yes : Choose Value : PBI Data → Send mail

No : Choose Value | Condition | Data Value → Send mail }

Case 2 : If the target value is hit, send email to Client.

{ Select visual ⟶ [...] ⟶ Manage alerts ⟶ Alert title : ⟶ Save & Close }
Alert rule:
Condition: Threshold :
Notification frequency :

{ Use Power Automate ⟶ ⊕ ⟶ Alert id ⟶ ⊕ Action ⟶ Send an email. ⟶ Save }

Pagenated Reports :—

→ Created as the purpose of printing.

{ BI Services ⟶ New ⟶ Paginated Report ⟶ (Select data models) ⟶ (Editor) ⟶ Save ⟶ Export }

Download PowerBI Report Builder
Install it.
Log in.

Deployment Pipeline :—

SDE process:

Developers → Testers. → Production.

Testers. ↓ Stakeholder

PowerBI → Data Source → Testing. → Production.

1 2 3 4 5

Many Developers.

Testing.
- Performance
- Optimization
- Load time
- Refresh time
- Auto Refresh

$\underbrace{\text{Developers} \rightarrow \text{Data Source} \rightarrow \text{Testing} \rightarrow \text{Production}}_{\text{Deployment Pipeline.}}$

Deployment Pipeline.

Developer → Test → Production.

{ BI Services → (Choose Workspace) → ... → Deployment Pipelines → New Pipeline → +Add → Create }

eg. Dep. Pipeline
↳ Premium Per User workspace.

{ Developement
↓
Assign Workspace
↓
Deploy. → Test ↓ ☐ ↓ Deploy → Production ↓ ☐ }

A copy is made.

A copy of testdata is made.

SQL Server ⟶ Semantic Report 1 ⟶ Report 1

SQL Server ⟶ Semantic Report 2 ⟶ Report 2

Same SQL Server.

(if 8 Refresh then server have to be hit 16 times.)

Problems: —

- Slow Server
- PowerBI Slow
- Inconsistent Report.
- Redundant Data Processing.
- Slow server leads to slow data transfer | processing for others as well.

SQL Server ⟶ Semantic Report 1 ⟶ Report 1

SQL Server ⟶ Semantic Report 2 ⟶ Report 2

Solution ⟶ Dataflow feature.

SQL Server ⟶ Data Flow ⟶ Semantic Report 1 ⟶ Report 1

Data Flow ⟶ Semantic Report 2 ⟶ Report 2

(Server is hit 8 times.)

(Dataflow is hit 16 times)

(Cache Memory)

It also optimizes Power BI.

Creating Data Flow

{ BI Services → Workspace → +New item → Dataflow }

{ Add new tables → (Connect SQL Server) → Next }

{ BI Desktop → Transform data → New Source → More → Dataflows → Connect. }

Connecting tables. to dataflow

{ BI Desktop → Transform data → (Select table) → Advance Editor → (Copy Code) →

→ (Select table) → Source → (Paste Code) }

Pagenated Reports :—

→ Created as the purpose of printing.

{ BI Services → New → Paginated Report → (Select data models) → (Editor) → Save → Export }

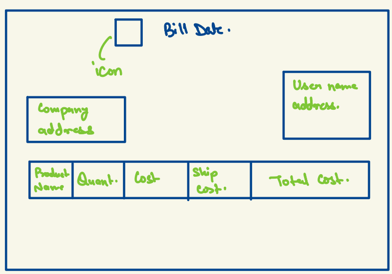
Download PowerBI Report Builder

Install it.

Log in.

(Continued)

Case Study : Customer bring invoice no. , Global Superstore provides bill.



{ Data Source → Add PowerBI Semantic Model → (Select Reports Model) }

{ Datasets → Add Dataset → (Name) (DataSource :) (Query designer) ↓click. (Choose table & data you need) → ok }

* Create new data set for parameters

* Don't use same dataset columns.

{ Datasets $\xrightarrow{\text{R.Click.}}$ (Name) → (Query Designer) }

{ Parameters → Add parameters → (Name) (Prompt : <Message>) → Available Values

→ Get values from query. → (Dataset) → (Value field) → (Label field) → OK }